

Assigning Diagonal of square to D1, and convert sins to values.

$$\frac{0.707106781}{0.866025405} = \frac{D2}{2 * \sqrt{2}}$$

multiply both sides by $2 * \sqrt{2}$

$$\frac{2}{0.866025405} = D2$$

$$D2 = 2.309401076$$

Observation

if i multiply $\cos(45)$ by 4, i get the diagonal of square. with side 2

$$\cos(45) = \frac{\sqrt{2}}{2}$$

$$4 * \cos(45) = 4 * \frac{\sqrt{2}}{2}$$